

Torque Control Method for Permanent Magnet Synchronous Motor Operating in Field Weakening Region at Middle Speed Range

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In this paper, a torque control method for permanent magnet synchronous motors (PMSMs) functioning in the middle speed range of the field weakening control region is proposed. The current resonance due to d axis and q axis coupling cannot be neglected. The current resonance is depressed by the appropriate design of PID controller for the q-axis current. The proposed controller designing method is verified both numerically and experimentally. The influence of parameter changes is also examined to reveal that the L_q change affect the torque performance more than L_d and R_m does.

Keywords: PMSM, field weakening control, torque control, PID compensator

1. Introduction

Permanent magnet synchronous motors (PMSMs) have been used in various application fields, such as railway vehicle tractions, automotives, air conditioners, elevators, etc., because the efficiency of PMSMs is higher than the one of induction motors. PMSMs are driven with the field weakening control when the terminal voltage of PMSMs reaches to the maximum output voltage of inverters.

Several field weakening control methods for PMSMs have been proposed so far⁽¹⁾. In reference (2) and (3), the current vector direction to weaken the field is changed depending on the rotor speed to maximize the motor torque and current ratio. The torque control response is not discussed in both (2) and (3). In reference (4), a voltage control pattern to minimize torque step change time for the fastest response is derived based on the Lagrange optimization method. However, the voltage pattern is entirely feedforward control system. Thus the parameter changes may affect much the torque response. Since a method proposed in reference (5) is based on field weakening method, the same problems occur with the parameter changes. In reference (6), the error of the phase RMS current and its reference is compensated by the voltage phase angle in the voltage limit range. Using the phase RMS current is effective to limit the current amplitude. However the absolute torque is not controlled accurately in this method.

The d-axis and q-axis coupling is a problem for higher torque control performance in PMSMs. In the field weakening region, only the voltage phase can be manipulated if the inverter outputs its maximum voltage. In this case, decoupling control over d-axis and q-axis is impossible because only a current can be controlled. Feedback control of q-axis current is one of effective measures to control the torque ac-

curately, because the relation between the q-axis current and the torque is more linear than the case using the phase RMS current⁽⁷⁾. Thus, the strategy of the q-axis current feedback to manipulate the phase angle of the voltage vector in the middle and higher speed range is proposed⁽⁸⁾. In the control method, the error of q-axis currents is compensated by an integration compensator, which is featured by a simpler controller designing procedure. However, the control method using the integration compensator can be effective only in the higher speed range, because the integrator cannot depress the affection of current oscillations caused by the coupling over d-axis and q-axis. Thus the method is available only in the case that the armature reaction reactance can be neglected due to the higher rotor frequency than the cut off frequency of the stator circuit. To cope with the problem in the middle speed range, a control method is proposed to apply PID compensator to the q-axis currents controller, instead of integrator⁽⁹⁾. The PID compensator is designed to reduce the peak gain of the PMSM impedance, which is caused by the d-axis and q-axis coupling. Thus the controller can decouple the PMSM current control system even in the middle speed range. This feature is preferable for the mobile traction applications such as railway vehicle tractions and automobile tractions, because the wider field weakening range is required to save the inverter and the motor size and their mass in these applications. The characteristics of the proposed control method in (9) are discussed further in this paper.

This paper starts with a numerical simulation to study current oscillation phenomenon due to the coupling over d-axis and q-axis systems. The characteristics of the control system are analyzed to reveal the current oscillation phenomenon. To design the PID compensator, a linearized model of the PMSM is derived, not neglecting the phase lag characteristics due to the d-axis inductance and q-axis inductance in the stator windings. Then a controller is designed so that the open-loop frequency characteristics of the control system are regarded as an integral element. The results simplify the design of frequency characteristics of the control system. The

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PID compensator is adopted depending on the rotor angular frequency. This compensator design provides the torque control performance independent from the rotor frequency. In the last part of this paper, the proposed designing method of the compensator gains is examined by the step response of the torque (q-axis) current to its reference with experimental tests with 1.0 kW class interior permanent magnet synchronous motors (IPMSMs). Torque step responses at different angular frequencies are verified as results. In addition, the torque control performance is theoretically and experimentally verified in the case that motor parameters such as resistance and inductances of stator windings change. It is revealed that the q-axis inductance cause influences the torque step performance. This paper contributes to provide a higher torque performance for the wider field weakening range suitable for the mobile traction applications.

2. Field Weakening Control

2.1 Control Method in Field Weakening Range In the speed range where the terminal voltage of the PMSM reaches to the limit of the inverter output voltage, PMSM is controlled with the field weakening control. In this speed range, the terminal voltage V_m is high enough to neglect stator resistance drop. Thus, V_m can be expressed by (1).

Figure 1 shows the voltage vectors and the current vectors in the field weakening range. From Fig.1, d-axis and q-axis current references can be expressed as shown in (2), (3). By substituting (2) and (3) to (1), the armature current vector phase angle δ^* can be obtained as (4) when the voltage reaches its maximum value V_{mmax} .

$$V_m = \omega \sqrt{(-L_q I_q)^2 + (\Phi_f + L_d I_d)^2} \dots\dots\dots (1)$$

$$I_d^* = -I_m \sin \delta^* \dots\dots\dots (2)$$

$$I_q^* = I_m \cos \delta^* \dots \dots \dots (3)$$

$$\delta^* = \sin^{-1} \frac{L_d \Phi_f - \sqrt{(L_d \Phi_f)^2 + (L_d^2 - L_q^2) \left(\frac{V_{m \max}^2}{\omega^2} - L_q^2 I_m^2 - \Phi_f^2 \right)}}{I_m (L_d^2 - L_q^2)} \quad (4)$$

2.2 Linearity of Torque Control Characteristic

Thus, it is more effective that the voltage vector phase angle change compensates the error of I_q and I_q^* so as to control torque rapidly and stably.

Figure 2 shows the field weakening control system which

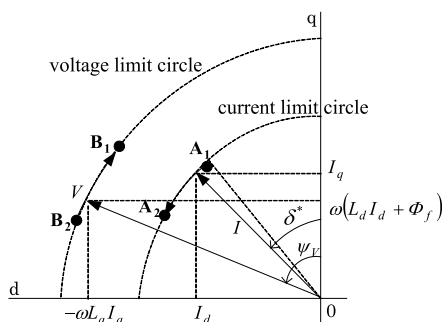


Fig. 1. Vector diagram of the voltage vector phase angle control

compensates the error between the torque current I_q and its reference I_q^* . From (2), (3) and (4), the current references I_d^* and I_q^* can be calculated when V_m is set its maximum value.

The voltage references V_d^* and V_q^* can be calculated by the current references, the PMSM constant parameters and the rotor frequency. A feedforward component of the voltage vector phase angle φ_{vFF} can be calculated by the voltage references. Then the voltage vector amplitude is limited to $V_{m\max}$.

$$\begin{pmatrix} V_d^* \\ V_q^* \end{pmatrix} = V_{m\max} \begin{pmatrix} -\sin\varphi_{vFF} \\ \cos\varphi_{vFF} \end{pmatrix} \dots\dots\dots (5)$$

In addition, according to (6), a feedback component of the voltage vector phase angle φ_{VFB} can be calculated. Where $C_{PID}(s)$ is a PID compensator expressed by (7) and K_{phP} , K_{phI} , K_{phD} are gains for a proportional element, an integrator element and differential element, respectively. New voltage vector references V_d^{**} and V_q^{**} obtained as (8) by changing coordinate.

$$\varphi_{vFB} = C_{PID}(s) (I_q^* - I_q) \dots\dots\dots (6)$$

$$C_{PID}(s) = K_{phP} + K_{phI}/s + K_{phD} \cdot s \dots\dots\dots (7)$$

$$\begin{pmatrix} V_d^{**} \\ V_q^{**} \end{pmatrix} = V_{m \max} \begin{pmatrix} \cos \varphi_{vFB} & -\sin \varphi_{vFB} \\ \sin \varphi_{vFB} & \cos \varphi_{vFB} \end{pmatrix} \begin{pmatrix} V_d^* \\ V_q^* \end{pmatrix} \dots\dots\dots (8)$$

3. Design of the Compensator Gains

3.1 Characteristics of the Proposed Control Method

The block diagram of the proposed torque current control system is shown in Fig. 3. Figure 4 shows the simplified block diagram of the proposed control system. In Fig. 4, the PID compensator gains can be analytically designed by neglecting feedforward control block $G_{FF}(s)$. In this section, $G_{PMSM}(s)$, the transfer function of PMSM model from φ_{vFB} to I_q , is derived for the analytical design of the PID compensator. If a feedback component of the voltage vector phase angle φ_{vFB} is small enough, (8) is approximated to (9). Where $\cos \varphi_{vFB}$ and $\sin \varphi_{vFB}$ are approximated by $\cos \varphi_{vFB} \approx 1$ and $\sin \varphi_{vFB} \approx \varphi_{vFB} \cdot V_{d_ave}^*$ and $V_{q_ave}^*$ are the equilibrium point of the voltage references under the approximated linearization. $V_{d_ave}^*$ and $V_{a_ave}^*$ are expressed by (10) and (11).

From (10) and (11), speed electromotive force $\omega_{ave}\Phi_f$ by a permanent magnet flux is dominant component in $V_{q_{ave}}^*$. Since $V_{q_{ave}}^*$ is much larger than $V_{d_{ave}}^*$, $V_{d_{ave}}^*$ can be neglected.

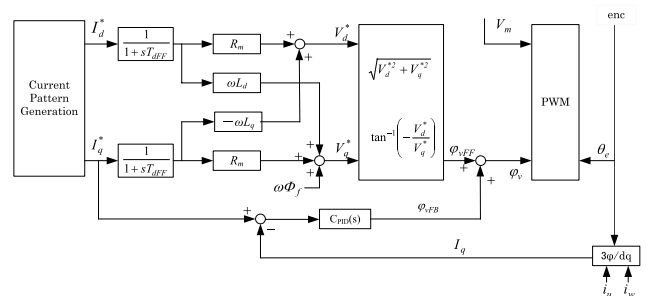


Fig.2. Block diagram of field weakening control of PMSM/with q-axis current feedback

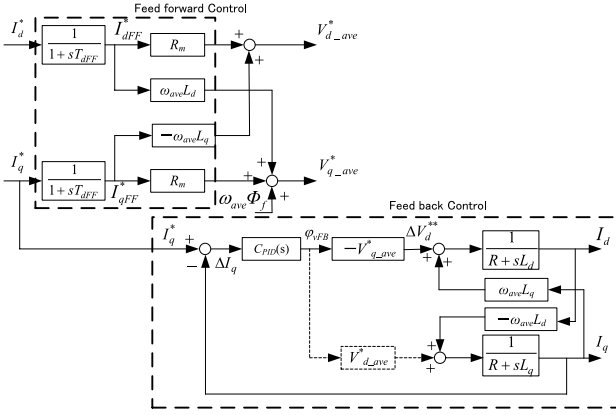


Fig. 3. Block diagram of the proposed control system

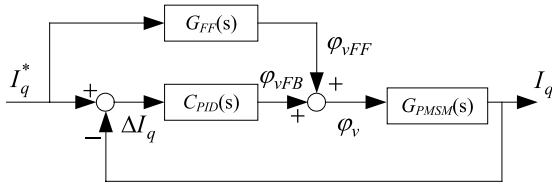


Fig. 4. Simplified block diagram of the control system

Table 1. Constant parameters of PMSM

Item	Constants
Stator coil resistance R_m	1.0[Ω]
Constants of e.m.f. ϕ_f	0.21[Vs]
D axis inductance L_d	12.0[mH]
Q axis inductance L_q	14.0[mH]
Pole	8
Inertial load J_m	0.76[kgm]
Control period	2.0 [ms]

$$\begin{pmatrix} \Delta V_d^{**} \\ \Delta V_q^{**} \end{pmatrix} = \begin{pmatrix} 0 & -\varphi_{vFB} \\ \varphi_{vFB} & 0 \end{pmatrix} \begin{pmatrix} V_{d_ave} \\ V_{q_ave} \end{pmatrix} \quad (9)$$

$$V_{d_ave}^* = R_m I_{dFF}^* - \omega_{ave} L_q I_{qFF}^* \quad (10)$$

$$V_{q_ave}^* = R_m I_{qFF}^* + \omega_{ave} (L_d I_{dFF}^* + \Phi_f) \quad (11)$$

In Fig. 4, $G'_{PMSM}(s)$ can be expressed as (12) by (9) and (11).

$$G'_{PMSM}(s) = \frac{\omega_{ave} V_{q_ave} L_d}{L_d L_q s^2 + R_m (L_d + L_q) s + (R_m^2 + \omega_{ave}^2 L_d L_q)} \quad (12)$$

If $\omega_{ave} \Phi_f$ is much larger than $\omega_{ave} L_d I_d$ in (12), $V_{q_ave}^*$ can be approximated by (13).

$$V_{q_ave} \cong \omega_{ave} \Phi_f \quad (13)$$

The inequality expression (14) is obtained.

$$R_m^2 \ll \omega_{ave}^2 L_d L_q \quad (14)$$

From (13) and (14), $G'_{PMSM}(s)$ is regarded as the form of (15).

$$G_{PMSM}(s) = \frac{\omega_{ave}^2 L_d \Phi_f}{L_d L_q s^2 + R_m (L_d + L_q) s + \omega_{ave}^2 L_d L_q} \quad (15)$$

$G_{loop}(s)$, which is a transfer function of the open-loop from I_q^* to I_q in Fig. 4, can be expressed by (16). $G_{close}(s)$, which is

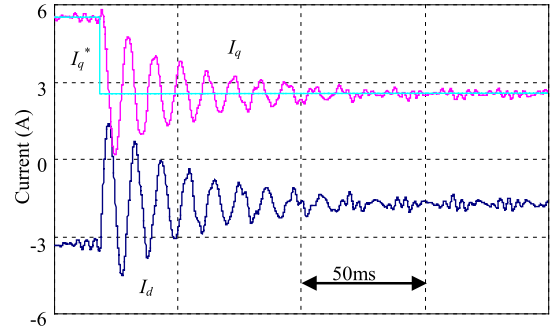
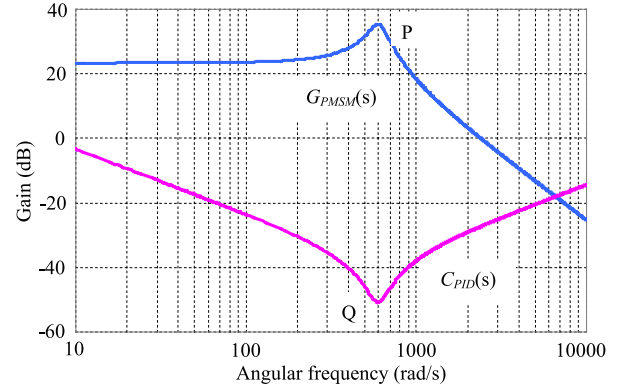


Fig. 5. Current oscillations phenomenon by numerical simulation


 Fig. 6. Frequency characteristics of $G_{PMSM}(s)$ and $C_{PID}(s)$

a transfer function of the closed-loop from I_q^* to I_q in Fig. 4, can be expressed by (17).

$$G_{loop}(s) = C_{PID}(s) \cdot G_{PMSM}(s) \quad (16)$$

$$G_{close}(s) = \frac{G_{loop}(s)}{1 + G_{loop}(s)} \quad (17)$$

From Fig. 3, rapid change of φ_{vFB} causes current oscillations due to speed electromotive forces interfering between d-axis and q-axis. In Fig. 5, this phenomenon is shown by a numerical calculation, where angular frequency of PMSM ω is 600 rad/s. The PMSM parameters used in the numerical simulation are shown in Table 1. Figure 5 shows that angular frequency of the current oscillations is almost equal to angular frequency of PMSM.

The frequency characteristic of $G_{PMSM}(s)$ is shown in Fig. 6, where ω is 600 rad/s. Figure 6 shows that the angular frequency where the peak gain of $G_{PMSM}(s)$ appears is almost equal to angular frequency of PMSM. Thus, the current oscillations can be depressed by cancelling the peak gain at the rotor angular frequency.

3.2 The Design of Compensator Gains As shown in Fig. 6, P stands for a point that gives the peak gain of $G_{PMSM}(s)$ and Q stands for a point that gives the bottom gain of $C_{PID}(s)$. $C_{PID}(s)$ stands for a PID compensator and is a transfer function of $\Delta I_q / \varphi_{vFB}$. $C_{PID}(s)$ is designed to cancel the peak gain of $G_{PMSM}(s)$, by its bottom gain.

The frequency characteristics of the closed-loop transfer function $G_{close}(s)$ is preferable to be the first-order lag element with the time constant of about 10 ms. Thus, the specification of $G_{loop}(s)$ must be set as an integral element with cut off angular frequency of 100 rad/s. The frequency

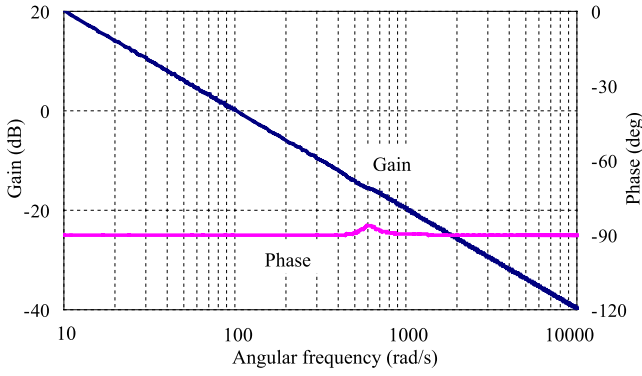


Fig. 7. Bode diagram of open-loop transfer function $G_{loop}(s)$

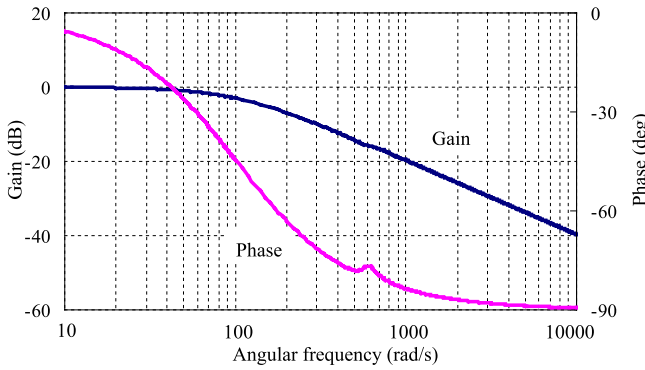


Fig. 8. Bode diagram of closed-loop transfer function $G_{close}(s)$

characteristic of $C_{PID}(s)$ is designed to cancel the peak gain of $G_{PMSM}(s)$ and to have cut off angular frequency at 100 rad/s, as shown in Fig. 8.

The time constant T_{dFF} of the feedforward control in Fig. 3 is determined before designing the PID compensator. This control method applies low-pass filters to avoid current oscillations due to the feedforward components of the voltage references. Thus, T_{dFF} would be better to be set about 10 ms or more to avoid the peak gain of $G_{PMSM}(s)$. In addition, T_{dFF} is set larger than the time constant of the I_q step response, so that the affection of $G_{FF}(s)$ may be neglected for the simplify design procedure. Thus, T_{dFF} is set 30 ms.

The feedback controller is designed as followings. The angular frequencies and gains at point P and Q can be analytically derived as follows. (15) is expressed by (18).

$$G_{PMSM}(s) = \frac{a_0}{b_2 s^2 + b_1 s + b_0} \quad \dots\dots\dots (18)$$

Where, a_0 , b_0 , b_1 and b_2 are,

$$\begin{aligned} a_0 &= \omega_{ave}^2 L_d \Phi_f \\ b_0 &= \omega_{ave}^2 L_d L_q, \quad b_1 = R_m(L_d + L_q), \quad b_2 = L_d L_q \end{aligned}$$

Thus, gain of $G_{PMSM}(s)$ can be expressed by (19).

$$20 \log_{10} |G_{PMSM}(j\omega)| = 20 \log_{10} \sqrt{\frac{a_0^2}{b_2^2 \omega^4 + (b_1^2 - 2b_0 b_2) \omega^2 + b_0^2}} \quad \dots\dots\dots (19)$$

Therefore, the angular frequency ω_P which gives peak gain of $G_{PMSM}(s)$, ω_P is calculated by (20).

$$|G_{PMSM}(j\omega)| = \sqrt{\frac{a_0^2}{b_2^2 \omega^4 + (b_1^2 - 2b_0 b_2) \omega^2 + b_0^2}} \quad \dots\dots\dots (20)$$

ω_P can be defined as the angular frequency which gives the minimum value of (21).

$$\frac{1}{|G_{PMSM}(j\omega)|^2} = \frac{1}{a_0^2} \{b_2^2 \omega^4 + (b_1^2 - 2b_0 b_2) \omega^2 + b_0^2\} \quad \dots\dots\dots (21)$$

When ω_P^2 is represented by K , (22) is obtained by differentiating (21) by K .

$$\frac{d}{dK} \left\{ \frac{1}{|G_{PMSM}(j\omega)|^2} \right\} = \frac{1}{a_0^2} \{2b_2^2 K + (b_1^2 - 2b_0 b_2)\} = 0 \quad \dots\dots\dots (22)$$

Consequently, ω_P can be given by (23).

$$\omega_P = \sqrt{K} = \sqrt{b_0/b_2 - (b_1/\sqrt{2}b_2)^2} \quad \dots\dots\dots (23)$$

By substituting ω_P in (23) into (19), the peak gain of $G_{PMSM}(s)$ g_P can be calculated. The gain of the transfer function of the PID compensator in the frequency range is expressed by (24).

$$|C_{PID}(j\omega)| = \sqrt{K_{phP}^2 + (K_{phD}\omega - K_{phI}/\omega)^2} \quad \dots\dots\dots (24)$$

Therefore, ω_M can be obtained by differentiating (24) by ω and setting (25) as zero.

$$\frac{d}{d\omega} |C_{PID}(j\omega)| = \frac{2K_{phD}\omega_M - 2K_{phI}/\omega_M^3}{\sqrt{K_{phP}^2 + (K_{phD}\omega_M - K_{phI}/\omega_M)^2}} = 0 \quad \dots\dots\dots (25)$$

Consequently, ω_M can be given by (26).

$$\omega_M = \sqrt{K_{phI}/K_{phD}} \quad \dots\dots\dots (26)$$

By substituting ω_M into (19), g_M of the minimum value of the $C_{PID}(s)$ can be expressed by (27).

$$g_M = 20 \log_{10} K_{phP} \quad \dots\dots\dots (27)$$

ω_M stands for the cut off angular frequency of the PID compensator is designed to be equal to ω_P , as expressed by (28).

$$\omega_M = \omega_P \quad \dots\dots\dots (28)$$

The gain of $G_{loop}(j\omega_P)$ at can be expressed by (29).

$$g_P + g_M = 20 \log_{10} (\omega_P/\omega_C) \quad \dots\dots\dots (29)$$

The gain of $G_{loop}(s) = G_{PID}(s) G_{PMSM}(s)$ at the cut off angular frequency ω_C is set 0 dB, as shown in (30).

$$20 \log_{10} |C_{PID}(j\omega_C) G_{PMSM}(j\omega_C)| = 0 \quad \dots\dots\dots (30)$$

From (28), (29) and (30), the gains of the PID compensator K_{phP} , K_{phI} and K_{phD} can be obtained as shown in (31), (32)

and (33).

$$K_{phD} = \frac{K_{phI}}{\omega_p^2} \dots \dots \dots (31)$$

$$\log_{10} K_{phP} = \log_{10} \left(\frac{\omega_p}{\omega_c} \right) - \frac{g_p}{20} \dots \dots \dots (32)$$

$$K_{phI} = \sqrt{\frac{b_2^2 \omega^4 + (b_1^2 - 2b_0 b_2) \omega^2 + b_0^2 - a_0^2 K_{phP}^2}{(\omega_c / \omega_p^2 - 1 / \omega_c)^2}} \dots \dots \dots (33)$$

These gains include angular frequency ω . Thus, these gains are adopted by ω to keep the control characteristics

4. Experimental Validation of Proposed Method

The results of step response of the q-axis current by the numerical simulations and the experimental tests are shown in Fig. 9 and Fig. 10, respectively. The PMSM parameters for this study are used in the Table 1. The step responses of q-axis current are examined at 600 rad/s and 800 rad/s to evaluate the affection of the rotor speed. In the both cases of the rotor frequencies, voltage vector phase angle φ_{vFB} of the feedback component enhance the motor torque.

Step responses of I_q and motor torque T_m show the first-order lag with the designed time constant of 10ms and they are independent from the affection of the different rotor frequency. The proposed control system and its designing method are verified.

5. Analysis of Parameter Changes of PMSM

As shown in (31)–(33), the nominal values, R_m , L_d and L_q , are used to design the controller gains. Thus, by the motor parameters change, the control performance can be deteriorated. The torque step responses at two cases of the temperatures such as -20 centi degrees and 120 centi degrees are examined, when the nominal value $R_{Nominal}$ is $1.0 [\Omega]$ at 20 centi degrees and temperature coefficient α is $0.0393 [\Omega/\text{deg}]$.

In Fig. 10(I)(a), the peak gain of the transfer function of PMSM is influenced by the R_m change. However, Fig. 10(I)(b) shows that the cutoff angular frequency of the open-loop gain is independent from the change of R_m . In Fig. 10(I)(c), the closed loop gain is not affected by the change of R_m in lower frequency range where the gain is higher. Thus, change of the stator windings resistance R_m is not major problem for the torque control performance of the proposed control system.

50% and 150% of the nominal values of L_d are assumed in order to examine the influences of the d-axis inductance L_d change. The nominal value of d-axis inductance is $12.0 [\text{mH}]$. Though the peak gain of $G_{PMSM}(s)$ in 150% of the nominal value is higher than the nominal value in Fig. 10(II)(a), the cutoff angular frequency of the open-loop gain in Fig. 10(II)(b) is independent from the d-axis inductance changes. In Fig. 10(II)(c), the closed loop gain is not affected by the change of L_d in lower frequency range where the gain is higher as well as the case of R_m change. Thus, the L_d change also does not cause major problem on the torque control performance in the proposed control method.

However, as shown in Fig. 10(III)(a)–(c), change of the q-axis inductance L_q affects the torque step response more than

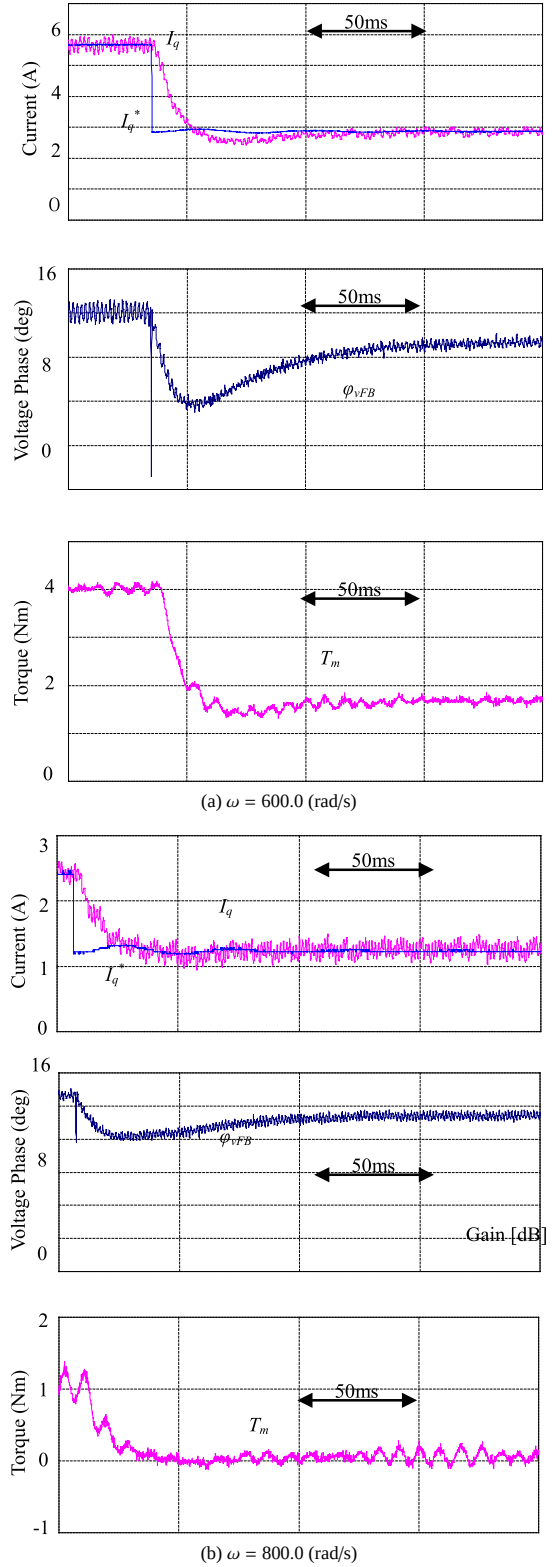


Fig. 9. Experimental results of the step response

the case of the R_m and L_d change. Figure 10(III) shows the gain of the transfer functions in cases of 50% and 150% of the nominal values of L_q . The nominal value of q-axis inductance is $14.0 [\text{mH}]$. The gains of PMSM transfer function $G_{PMSM}(s)$ and the open loop transfer function $G_{open}(s)$ changes due to the L_q change in all over frequency range. This causes the change of the dynamic range of the closed

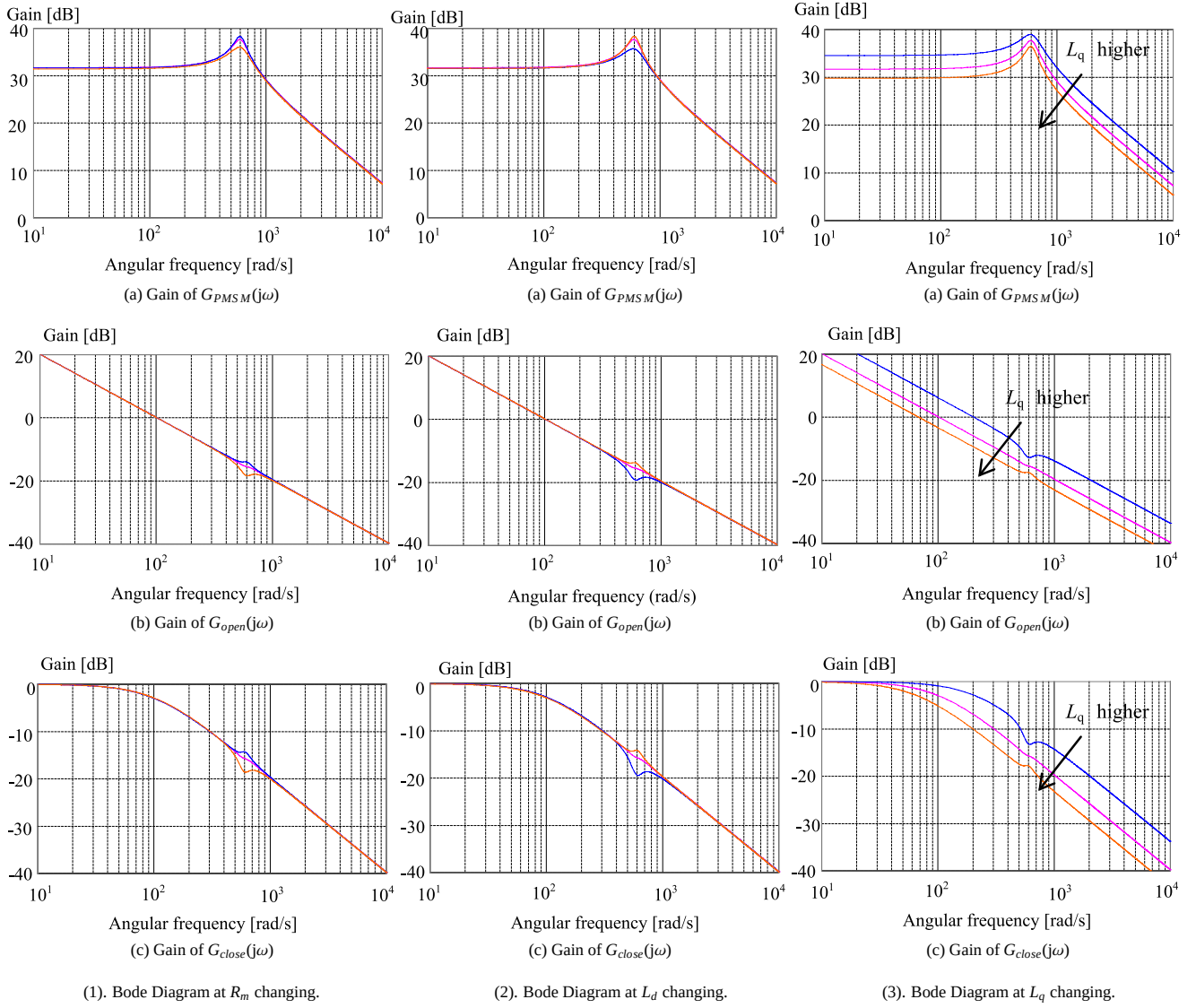


Fig. 10. Bode Diagram at parameters changing

loop function $G_{close}(s)$. Thus, the torque control performance is affected by the L_q change more than the case of the R_m and L_d change. This can be explained by (34) obtained from (19).

$$|G_{PMSM}(j\omega)| = \frac{1}{L_q} \sqrt{\frac{(\omega_{ave}^2 L_d \Phi_f)^2}{(L_d \omega^2)^2 + \{R_m^2 (L_d/L_q + 1)^2 - 2(\omega_{ave} L_d)^2\} \omega^2 + (\omega_{ave}^2 L_d)^2}} \quad \dots\dots\dots (34)$$

The d-axis inductance affects minor the gain of $G_{PMSM}(s)$ because L_d is almost cancelled both in the numerator and denominator of (34). The affection of the armature resistance changing is also weak because $R_m^2 (L_d/L_q + 1)^2$ is small enough to neglect the affection. However, the q-axis inductance affects more the gain of $G_{PMSM}(s)$, because the gain of the PMSM much depends on L_q as shown in (35), which can be converted from (34).

$$20 \log_{10} |G_{PMSM}(j\omega)| = -20 \log_{10} L_q + 20 \log_{10} A \quad \dots\dots\dots (35)$$

This is because only the q-axis current is controlled by the

d-axis voltage that consists mainly of the armature reaction voltage by q-axis inductance.

6. Verification of Parameter Changes of PMSM by Experimental Tests

The experimental tests validate the results of theoretical analysis in section 5. In the experimental tests, the values for the controller are set as 50% and 150% of the nominal values.

Figure 11 shows the experimental tests results of the step response of q-axis current at 600 rad/s.

Figure 11(a) shows the step response of the nominalized q-axis current. These are the cases of the 50% and 150% of nominal resistance of the stator windings R_m . The step response is almost independent from the R_m change.

Figure 11(b) shows the step response of nominalized q-axis current in the case of d-axis inductance changes. As shown in the gains characteristics of Fig. 10(II), the step response is affected minor by the L_d changes.

The q-axis inductance change causes major influence on the step response of the nominalized q-axis current as shown in Fig. 10(III). In the case of the lower value of L_q than the

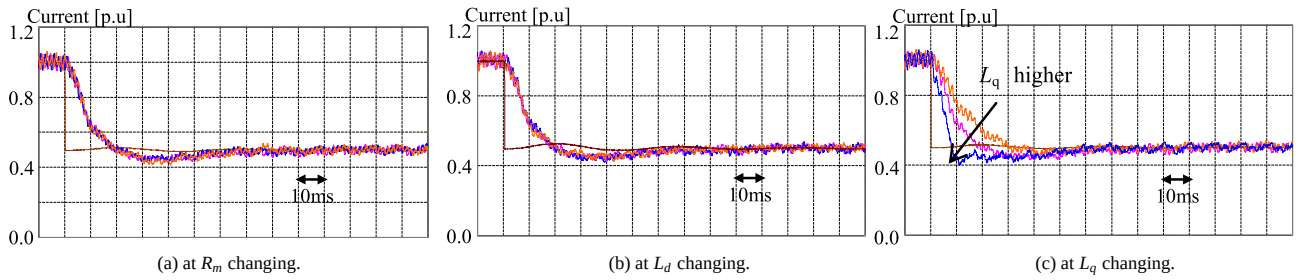


Fig. 11. Experimental results of step response of nominalized q-axis current at parameters changing

nominal value, the step response is slower and in the case of the higher value of L_q than the nominal value, the step response is faster but the over shoot increase. As far as the overshoot cause major problem in the whole control system, the value of the L_q in the PID compensator is set smaller than the range of the possible actual value of L_q .

7. Conclusion

In this paper, q-axis current control method with PID compensator in the field weakening range of PMSM are presented and examined its characteristics. The control system compensates the error between I_q and its reference I_q^* by manipulating the voltage vector phase angle φ_{vFB} . The proposed method is featured to be available in not only the higher speed range and but the middle speed range by coinciding the frequency which gives the peak gains of PID compensator to synchronous angular frequency of the PMSM. Experimental tests verify the proposed control method. In addition, the influence on the torque control performance by the motor parameters changes are examined both theoretically and experimentally. These result in the q-axis inductance is more sensitive to the torque control performance, but the value of q-axis inductance in the controller is set smaller as far as the over shoot does not causes major problem. This study reveals that the proposed control method with PID compensator is available enough to drive PMSM in wider operational speed range in the field weakening region of PMSM.

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